**APPENDIX 1**

# MARKING RUBRICS

| | Excellent (90-100%) | Good (80-89%) | Adequate (60-79%) | Poor (20-60%) | Lacking 20-0% |
|---|---|---|---|---|---|
| **Abstract/ Introduction/ Theory** | 1.Includes complete well-written information on the background to understand the subject of the experiment and its significance.<br>2.Summarizes what is already known on the subject.<br>3. Includes the question to be answered by the lab and states hypothesis that is based on research and/or sound reasoning, described in the background.<br>4. Includes the description on how the hypothesis is going to be tested.<br>5. Deep understanding of the subject as well as creativity is demonstrated. | 1.Includes strong introductory information, most major points provided.<br><br>2.Most important details in the summary provided.<br>3. Most important details of the hypothesis covered.<br>4. An idea on how to test the hypothesis is included.<br>5. Good understanding is demonstrated. | 1.Includes good introductory information, missing some points in the summary.<br><br>2.Hypothesis is missing some minor details.<br><br>3.No description on how the hypothesis is going to be tested is provided.<br><br>4. Appropriate understanding of the subject is demonstrated. | 1. Introductory information provided, missing some major points.<br>2. Important details of the hypothesis are missing.<br>3. No description on how the hypothesis is going to be tested provided.<br>4. Limited understanding of the subject is demonstrated. | 1. Part is missing or plagiarized.<br>OR<br>2.Little background information is present.<br>3.Incorrect information is provided.<br>4. For the hypothesis, important experimental details are missing, of the hypothesis is missing.<br>5. Poor understanding or no understanding of the subject is demonstrated. |
| **methodology** | Description (not a copy of a protocol!) is included, detailed enough so it could be repeated by another scientist. Description is not overloaded with unnecessary details. | Description included, some steps are vague or unclear OR instead of description, a step-by–step protocol is copied from manual. | The description gives generalities, enough for reader to understand how the experiment was conducted | Would be impossible to repeat the experiment, reader must guess at how the data was gathered or experiment conducted | Part is missing or plagiarized |

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| **Results** | Results and data are clearly recorded, so it is easy for the reader to see trends. Excellent graphs, tables or figures with appropriate names and labels, excellent information and understanding of results, excellent interpretation of data and trends while referring to figures and tables. | Results are clear, but trends are not obvious or there are minor errors in organization. Good figures, graphs, or tables, good information results, good understanding of results, but some figure/table names or labels are missing. | Results are unclear, trends are not obvious, disorganized, but there is enough data to show the experiment was conducted. Appropriate figures, graphs or tables or drawings, some labels and figure names are missing. Only some understanding of results is evident. | Results are disorganized or poorly recorded, do not make sense; not enough data was taken to justify results. Missing or incorrect information. | Part is missing or plagiarized. Misleading, confusing and/or incorrect information. Poorly presented results. |
| **Discussion/ Conclusions** | 1.Summarizes data used to draw conclusions<br>2.Conclusions follow data (not wild guesses or leaps of logic)<br>3. Hypothesis is rejected or accepted based on the data.<br>4. Discussion of applications or realworld connections is present.<br>5. Deep understanding of the subject as well as creativity is demonstrated. | Good conclusion, important information provided, or one point from Excellent is missing | Appropriate conclusion. Two points from Excellent are missing. | Limited conclusion. Important information is missing. | Part is missing or plagiarized.<br>OR<br>No conclusion is made. Missing important information. Little effort or reflection is shown. |
| **References** | All references and citations are correctly written and present. More than two resources are cited. Student accurately cites to the sources throughout the text. | One reference or citation is missing or incorrectly present. | Appropriate citing and reference list. | References and citations are missing or errors detract significantly from paper. | Part is missing or plagiarized. Incorrect citing. |